PRIORITY QUEUES

Problem Solving with Computers-II





Emergency Room Triage

You're a software engineer at a hospital.
Patients arrive and need to be treated in order of severity.

1 = Critical (life-threatening)
2 = Urgent
3 = Moderate
4 = Minor

Emergency Room Triage

What if we used std::queue?

```
    t=1  Alice   (Moderate)   ← treated first
    t=2  Bob     (Critical)   ← waits behind Alice
    t=3  Carol   (Minor)
    ...
```

First come, first served — is that okay here?

C++ Priority Queue ADT

```
priority_queue<int> pq;

pq.push(20);
pq.push(20);
pq.push(80);
pq.push(50);
pq.push(100);

cout << pq.top();

pq.pop();
```

Live code ERTriage